PANKAJ PATEL

Full Stack Dev · Backend-Focused · MERN Stack

Ahmedabad, Gujarat · 9173731916 · ppankajpatel348@gmail.com · LinkedIn · GitHub

TECHNICAL SKILLS

Languages: JavaScript (primary), TypeScript, Java, Python, C
Backend: Node.js, Express.js, REST API Design, JWT Authentication, RBAC, Better-Auth, Google OAuth2
AI / ML: LangChain, Amazon Bedrock, Pinecone, Groq API, RAG Pipelines, Model Context Protocol (MCP)
Databases & ORM: MongoDB, MySQL, PostgreSQL, Prisma ORM
Caching & Queues: Redis, BullMQ, Inngest
Frontend: React.js, Next.js, TanStack Query, Tailwind CSS, HTML, CSS
DevOps & Tools: Docker, Git, GitHub, Postman, Zod
Learning: AWS (EC2, S3, IAM, RDS, CloudWatch, VPC), Model Context Protocol (MCP)

PROJECTS

AI-Powered Codebase RAG Assistant | *Node.js · Express · React · Amazon Bedrock · Pinecone · Amazon Titan · GitHub API* 2026
GitHub · Live Demo

- Built a full-stack application that allows developers to chat with their codebase, providing context-aware explanations and code analysis.
- Implemented semantic search by generating 1024-dimensional embeddings using Amazon Titan v2 and indexing them in Pinecone Cloud, enabling sub-second retrieval across large repositories.
- Designed a chunking engine that preserves structural code integrity by identifying logical boundaries (classes, methods, functions) for JavaScript, Python, and Java.
- Integrated vector similarity search with strict repository filtering and custom reranking to ensure precise context retrieval for complex developer queries.
- Implemented automated repository tree indexing to provide the LLM with a comprehensive overview of project hierarchy and cross-file relationships.

StudyMate – AI-Powered Personal Learning Assistant | *Next.js · React · TypeScript · Prisma · PostgreSQL · Pinecone · AWS Bedrock · Inngest* 2026
GitHub · Live Demo

- Architected an event-driven Next.js personal learning platform featuring Inngest background job queues for processing multi-page document uploads asynchronously.
- Engineered a high-throughput RAG document ingestion pipeline using LangChain, Amazon Bedrock embeddings, and vector namespaces in Pinecone.
- Integrated Better-Auth and Prisma ORM over PostgreSQL to enforce robust user authentication, session security, and dynamic database schemas.
- Developed an interactive AI learning suite incorporating dynamic quiz generation, automated slide presentation decks, concept visualization with Mermaid.js, and Upstash Redis rate limiting.

AI Gmail Assistant using MCP | *Node.js · Express.js · React.js · Model Context Protocol (MCP) · Groq API · Google OAuth2* 2026
GitHub

- Developed an agentic AI email assistant leveraging Model Context Protocol (MCP) and Groq LLM architecture to enable natural language inbox automation.
- Integrated Google OAuth2 authentication and Gmail API, empowering the AI agent to securely read unread messages, perform targeted search queries, and draft/send emails.
- Implemented custom MCP tool handlers with strict Zod schema validation (read_emails, search_emails, send_email), guaranteeing type-safe execution and robust error handling.
- Designed a responsive React + Tailwind CSS chat interface supporting multi-turn conversation memory and real-time step-by-step tool execution feedback.

EDUCATION

Master of Computer Applications (MCA) — L.D. College of Engineering, Ahmedabad 2024 – 2026

- CGPA: 8.41 · Top 5 rank in Semester 2 · Coursework: DSA, DBMS, OS, Software Engineering

Bachelor of Computer Applications (BCA) — V.S. Patel College of Arts and Science, Navsari 2022 – 2024

ACHIEVEMENTS & INTERESTS

- Ranked in Top 5 students in MCA Semester 2 at L.D. College of Engineering.
- Core interests: Backend Development, API Design, Scalable Systems, Generative AI & MCP, Cloud Infrastructure.